Fiscal Year:	FY 2020	**Task Last Updated:**	FY 01/23/2020
PI Name:	Long, Marshall Ph.D.		
Project Title:	Fuel Dilution Studies of Coflow Laminar Diffusion Flames in a Microgravity Environment		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	marshall.long@yale.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	203-432-4229
Organization Name:	Yale University		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	PO Box 208284		
PI Web Page:			
City:	New Haven	**State:**	CT
Zip Code:	06520-8284	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	03/01/2017	**End Date:**	08/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 8/31/2021 per D. Stocker/GRC and NSSC information (Ed., 4/20/2020) NOTE: Period of performance changed to 3/1/2017-2/28/2020 (original POP was 1/1/2017-12/31/2019) per D. Stocker/GRC (Ed., 1/17/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Smooke, Mitchell Ph.D. (Yale University)		
Grant/Contract No.:	NNX17AD49A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Coflow Laminar Diffusion Flame" grant NNX11AP43A, with investigators Drs. Marshall Long and Mitchell Smooke. Diffusion flames are the flame type of most practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in such systems is critical in the modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation, and radiative transfer take place. Turbulent diffusion flames are generally nonbuoyant due to the large flow velocities. In addition, in practical devices combustor configuration and flow field patterns may produce flames with sharp edges. Extinction along these edges can result from incomplete combustion. In other regions combustion generated soot particulates may be formed. To understand better the factors that affect flame extinction and particulate formation, we propose to focus this study on flame phenomena at the extremes of the fuel dilution spectrum – namely dilute flames near extinction and pure-fuel (sooting) flames. Previous work has shown that existing computational models are unable to predict accurately the behavior of these systems. The enhanced time and spatial scales in a microgravity environment will enable the stabilization of flames with increased levels of dilution and will help generate flames that have increased soot residence times compared to normal gravity facilities. This should provide an excellent platform to gain a better understanding of the factors that affect diffusion flame lift-off distance and extinction, and it should also provide an excellent test-bed to determine the roles of inception, surface growth, and oxidation processes in soot formation models. The overall goals of the research are to improve our understanding of the physical and chemical processes controlling diffusion (i.e., non-premixed) flame structure and lifting phenomena (i.e., stabilization) and to provide for rigorous testing of numerical models, including thermal radiation, soot formation, and detailed chemical kinetics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Combustion of fossil fuels is responsible for the majority of energy production in the world today and provides the power for transportation systems of all kinds. Effective modeling of combustion processes is critical to improving efficiencies and reducing pollutants in practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in diffusion flames will improve modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation, and radiative transfer take place.
Task Progress:	Analysis of Coflow Laminar Diffusion (CLD) Flame Data from Rounds One and Two With the end of two rounds of CLD Flame testing, efforts have turned to the analysis of more than 300 test points recorded. The images of methane and ethylene flames with varying levels of nitrogen dilution have been processed to characterize flame geometry and lift-off height. For methane flames, calculations of CH* concentrations have been made (CH* is responsible for the blue light emitted by these flames). Temperature profiles were generated from thin filament pyrometry (TFP) fiber sweeps on highly nitrogen-diluted flames of both methane and ethylene. Soot temperature and volume fraction distributions from sooting ethylene flames have also been calculated. Some observations on differences in these flames under normal and microgravity conditions are included below. Normal Gravity Experiments using the CLD Flame Burner for Comparison with Microgravity Results The Yale-based ground copy of the Advanced Combustion via Microgravity Experiments (ACME) CLD Flame experiment (including TFP fibers) was set up and matched as closely as possible to that on the International Space Station (ISS), to allow comparisons to be made between 1 g and 0 g results. This involved measurements on both methane and ethylene flames with varying levels of nitrogen dilution, selected to match closely the fuel concentrations and flow conditions of the microgravity data from the ISS experiments. Specific test cases investigated include soot free, highly nitrogen-diluted flames, and heavily sooting flames with low nitrogen dilution. Soot free methane flames were found for a dilution range between 30% to 60% methane (in all cases the balance of the fuel mixture is nitrogen). In normal gravity, fuel mixtures with fuel concentrations below 40% methane were found to be highly unstable and, in most cases, unsustainable. Therefore, flames with fuel mixture concentrations of 40% methane were chosen for direct comparison with the near-dilution-limit microgravity data. The coflow and fuel mixture velocities ranged from 10 to 30 cm/s at this concentration. Above 30 cm/s, these highly diluted flames experienced blow out. Ethylene flames possess far greater soot loading than methane flames. Therefore, ethylene flames with high fuel mixture concentrations were the focus of a heavily sooting flame investigation. The dilution ranges for these flames was performed from 40% to pure ethylene flames. The coflow and fuel mixture velocities ranged from 30 to 50 cm/s. Comparison of Normal and Microgravity Results Highly Diluted Methane Flames Flame images were captured under a large variety of flow conditions. Comparisons were made between microgravity and normal gravity highly diluted flames with 40% methane. Since the recorded images are integrated along the line of sight, they must be deconvoluted to provide cross-sectional data. This is typically done using a three-point Abel-inversion algorithm. Of the two gravity conditions, the flame lift-off height is more sensitive to increasing flow rate in normal gravity. Past 30 cm/s, the normal gravity flames are very unstable and highly susceptible to extinguishment. Microgravity flames are more stable and much higher flow rates and lift-off heights could be reached. Microgravity flames were captured with gas velocities up to 45 cm/s. It is also found that at this concentration, the microgravity flames are able to maintain a more curved cone shape while the normal gravity flames possess more of a curved disk shape. It was found from the flame images that the dilution limits of methane flames were extended in microgravity. From the ground-based copy of the CLD Flame experiment, it was found that 30% methane flames were unsustainable in normal gravity but were sustainable in microgravity. Full flame temperature profiles were constructed using TFP for 30% and 40% methane microgravity flames. A direct comparison of 40% methane flames in microgravity and normal gravity at 25 cm/s shows that the lifted characteristic of the normal gravity flame creates a flatter profile with a lower peak CH* mole fraction. Sooting Ethylene Flames Images were also captured of ethylene flames with fuel mixtures containing low dilution with nitrogen. These images were also deconvoluted using a three-point Abel-inversion algorithm. The deconvoluted data was then used to calculate the soot temperature using two-color ratio pyrometry and soot volume fraction using well known relationships of the optical properties of soot and an absolute light calibration. At lower fuel concentrations, the soot region possessed a

small-capped profile near the tip of the flame. But as the fuel concentration increases, the soot region elongates as the flame grows taller. The peak temperature also cools due to increased radiative cooling. With pure microgravity ethylene flames, the peak temperature transitions to the wings of the flame while the centers of the flame are found to be significantly cooler. The normal gravity flames never make a full transition to the wings under the fuel conditions investigated. The microgravity flames also grow taller faster than the normal gravity flames and peak temperature cools much quicker also. The structure of the soot volume fractions is similar to that of the soot temperature, where the maximum soot region possesses a capped profile at the lower fuel concentrations and transitions to the wings as the fuel concentration moves towards pure fuel. As the fuel mixture increases in ethylene concentration and the gas velocity increases, the soot volume fraction also increases. In all cases, the microgravity flames possess a much higher soot volume fraction than their normal gravity equivalents. In most cases, the microgravity flames are found to have a soot volume fraction with a factor of 6-8 times higher than the normal gravity flames.

Normal Gravity Experiments using the Standard Yale Coflow Burner for Verification of Processing of the ACME data

In addition to normal gravity tests on the ground copy of the ACME CLD Flame burner, our copy of the ACME color data camera was used for soot temperature and volume fraction measurements on the Yale standard coflow burner. Flames from this burner have well characterized temperatures and soot volume fractions, obtained over many years using multiple diagnostic techniques. Such testing allows for verification of calculations performed on the ACME experiments as well as characterization of the relative uncertainties and accuracies of measurements made with the ACME data camera. Specific issues investigated were the evaluation of the possibility of pixel cross-talk between color channels of the ACME color camera, most likely to occur when there is a large difference between signal intensities. Other tests were performed to evaluate the effect of the ACME color camera's lower pixel resolution, dynamic range, and signal/noise ratio compared to previously used digital cameras. Fiber measurements were also made with flames on the Yale standard burner. The light calibration from these were made as a confidence check of the performance between the ISS hardware and the ground-replica hardware. Additional ACME color data camera tests were performed on the standard flames, both with and without BG-7 color filters, to characterize differences in the measured quantities.

Improvements to Data Analysis

Since the beginning of ACME ISS testing, the CLD Flame team has made advancements in processing the ISS data as it was being generated. That work has continued this year with several notable advances. Evaluations were done of the FliPPID method (J.A.H. Dreyer et al. Appl. Opt., 58(10), pp. 2662-2670, 2019) as an alternative to conventional inverse Abel transforms. For some cases, this method has been shown to be useful for improving calculations of soot temperature and volume fraction done in axisymmetric flames. It is especially important for our case where the limited signal-to-noise ratio and spatial resolution of the ACME data camera result in undesirable artifacts near the centerline of the flame when inverted with other Abel transform methodologies. A drawback of the original version of the published code was that it is written in Python and execution was very slow. The code was rewritten in C++ with fitting-function options expanded, resulting in a factor of 300 speed increase and greater functionality. The improved routines have been incorporated into the oma2 program. As an aid to the other ACME experiments, an updated quick-start guide to installing the oma software and using it for processing data from the ACME color data camera was posted on GitHub. That repository also hosts the latest software versions of oma2 (for MacOS computers), oma2cam (which allows image capture using GigE and DSLR cameras), and QtOma2 (for Mac, Windows, and Linux operating systems). The repositories are available at http://github.com/ .

New Test Matrices

New test matrices were created for possible bonus testing in case there is a gap between ACME and subsequent Solid Fuel Ignition and Extinction (SoFIE) solid fuel combustion experiments. These experiments would concentrate on taking additional data on sooting flames, mismatched fuel/coflow velocities, continued evaluation of the light calibration of the color camera, and filling in some missing data in our previous tests.

Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Tinajero J, Giassi D, Stocker DP, Long MB. "Experimental Study on the Influence of Gravity on Highly Diluted and Sooting Coflow Flames." Presented at the 11th U.S. National Combustion Meeting, Pasadena, CA, March 24-27, 2019. 11th U.S. National Combustion Meeting, Pasadena, CA, March 24-27, 2019. https://wssci.us/meetings/ncm2019/ , Mar-2019
Abstracts for Journals and Proceedings	Tinajero J, Giassi D, Stocker DP, Long MB. "Experimental Study on Sooty Ethylene Coflow Flames in Microgravity." Oral presentation, 2019 MACCCR Fuel/Combustion Research Review (Multi-Agency Coordinating Committee for Combustion Research), Arlington, VA, September 9-12, 2019. 2019 MACCCR Fuel/Combustion Research Review, Arlington, VA, September 9-12, 2019. , Sep-2019
Abstracts for Journals and Proceedings	Tinajero J, Giassi D, Stocker DP, Long MB. "Experimental Study on Sooty Coflow Flames in Microgravity." Oral presentation, 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
Awards	Long MB. "Elected Fellow of the Combustion Institute, February 2019." Feb-2019